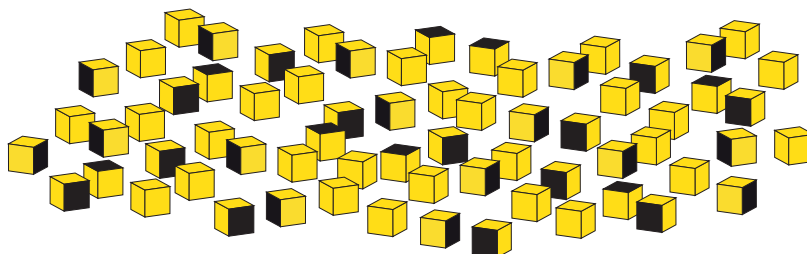


Examiner
only

- 8.** The atoms of radioactive substances decay at random.

- (a) Students are given 200 cubes each with one side coloured black. Describe how they would use the cubes to model radioactive decay **and** find the half-life.

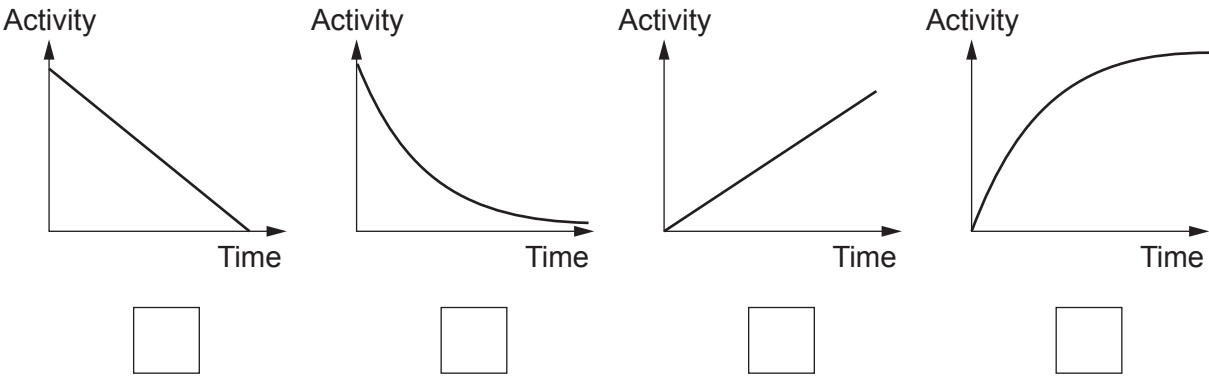
[6 QER]



(b) An activity-time graph can be drawn for a radioactive substance.

Place a **tick (✓)** in the box beneath the graph that shows how the activity of a radioactive substance changes with time.

[1]



(c) In an extension to their experiment, each group of students in a class is given 270 6-sided dice.
After each throw they remove the dice that show a ONE or a SIX facing upwards.

Their results are shown below.

Number of throws	Number of ONES removed	Number of SIXES removed	Number of dice remaining
0			270
1	40		185
2	37	30	118
3	18	17	
4	14	14	55
5	10	9	36

(i) After the **first throw**, how many ONES would you have **expected** to be removed? [1]

number of ONES expected

(ii) **Complete the table.** [2]

(iii) Why does combining results from several groups of students better model radioactive decay? [1]

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